

Homework 1

Ximena Ortiz Linear Algebra Sep 18, 2026 GRADE: 33/40 (Prof. R. Castellano)

Problem 1

(a)

First I scale u by 2 and subtract v entry by entry, then dot the result with w :

$$2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 8]^T$$

$$(2u - v) \cdot w = (3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 24$$

[ANSWER: 24]

[PROF: ✓]

(b)

Halving w first gives $(1/2)w = [-2, 1, 3]^T$, and adding that to v gives

$$v + (1/2)w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T.$$

The length is the square root of the sum of the squares of the entries:

$$\|v + (1/2)w\| = \sqrt{(-1)^2 + 5^2 + 1^2} = \sqrt{1 + 25 + 1} = \sqrt{27}$$

Since $27 = 9 \cdot 3$, this simplifies to $3\sqrt{3}$, which is about 5.20.

[ANSWER: $3\sqrt{3} \approx 5.20$]

[PROF: ✓]

(c)

$\cos \theta = (u \cdot w) / (\|u\| \|w\|)$, so I need the dot product and both lengths first.

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$\|u\| = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$\|w\| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$\cos \theta = 8 / (\sqrt{14} \cdot 2 \sqrt{14}) = 8 / (2 \cdot 14) = 8/28 = 2/7$$

The inverse cosine of $2/7$ in degrees is $73.398\dots^\circ$, which I round to two decimals.

[ANSWER: $\theta \approx 73.40^\circ$]

[PROF: ✓]

[PROF: Problem 1 5/5]

Problem 2

(a)

AB is undefined, because A is 3×3 and B is 2×3 , so the 3 columns of A do not match the

2 rows of B . BA is defined (2×3 times 3×3) and the product is 2×3 .

$$\text{row 1} = [1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$$

$$\text{row 2} = [3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$$

$$\text{[ANSWER: } AB \text{ undefined; } BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 4 & -1 \end{bmatrix}]$$

[PROF: ✓]

(b)

AV is defined (3×3 times 3×1) and the result is another 3×1 column:

$$AV = [2(1) + (-1)(-2) + 0(3), 1(1) + 3(-2) + (-2)(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$$

VA is undefined, because V is 3×1 and so it has only 1 column, while A has 3 rows.

$$\text{[ANSWER: } AV = [4, -11, -5]^T; VA \text{ undefined}]$$

[PROF: ✓]

(c)

Aw is undefined: A has 3 columns but w is 1×3 , so it has only 1 row, and $3 \neq 1$.

wA is defined (1×3 times 3×3) and the answer is a 1×3 row vector:

$$wA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)] = [4, -6, -1]$$

[ANSWER: Aw undefined; $wA = [4 \ -6 \ -1]$]

[PROF: ✓]

(d)

Both exist but have different shapes: vw is 3×1 times 1×3 , so 3×3 , and wv is 1×1 .

$$vw = [1(2) \ 1(0) \ 1(-1)] \ [2 \ 0 \ -1]$$

$$[-2(2) \ -2(0) \ -2(-1)] = [-4 \ 0 \ 2]$$

$$[3(2) \ 3(0) \ 3(-1)] \ [6 \ 0 \ -3]$$

$$wv = 2(1) + 0(-2) + (-1)(3) = 2 - 3 = -1, \text{ which is the dot product of } w \text{ and } v.$$

[ANSWER: vw as above; $wv = [-1]$]

[PROF: ✓]

(e)

A is square, so $A^2 = AA$ is defined and is again 3×3 . Multiplying row by column:

$$\text{row 1} = [2(2) + (-1)(1) + 0(0), 2(-1) + (-1)(3) + 0(4), 2(0) + (-1)(-2) + 0(1)] = [3, -5, 2]$$

$$\text{row 2} = [1(2) + 3(1) + (-2)(0), 1(-1) + 3(3) + (-2)(4), 1(0) + 3(-2) + (-2)(1)] = [5, 0, -8]$$

$$\text{row 3} = [0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 16, -7]$$

[ANSWER: $A^2 = [3 \ -5 \ 2]$

$$[5 \ 0 \ -8]$$

$$[4 \ 16 \ -7]]$$

[PROF: ✓]

(f)

B^2 is undefined: squaring needs the columns to match the rows, but B is 2×3 , so BB

would require $3 = 2$.

[ANSWER: B^2 undefined (B is not square)]

[PROF: ✓]

[PROF: Problem 2 6/6]

Problem 3

(a)

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

so $x_1 = 2, x_2 = -1, x_3 = 1$

[ANSWER: $x = [2, -1, 1]^T$]

[PROF: -1.5 E1 No elimination is shown; the solution is stated straight off the augmented matrix, and the problem requires every row operation.]

(b)

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & 1 & | & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$[0 \ 1 \ 3 \mid -2]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

$$R_1 \rightarrow R_1 - 2R_2: [1 \ 0 \ -7 \mid 8]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

The bottom row is all zeros, so it tells me nothing and I drop it. Taking $z = 0$, the first row gives $x = 8$ and the second gives $y = -2$, so the system has one solution.

[ANSWER: $x = 8, y = -2, z = 0$]

[PROF: -1 S7 The zero row means z is free, so the system has infinitely many solutions, not one.;

-1 E4 Setting $z = 0$ gives one particular solution, not the general solution in parametric form.]

[PROF: Problem 3 2.5/6]

Problem 4

(a)

Let x_1, x_2, x_3 be the masses in kilograms of Alloy 1, Alloy 2, Alloy 3 in the blend. The finished 100 kg batch is 36% copper, 44% zinc, 20% nickel, so it holds 36 kg of copper, 44 kg of zinc, 20 kg of nickel. No metal is lost, so for each metal the mass going in equals the mass coming out:

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

[PROF: ✓]

(b)

I multiplied every equation by 10 first so I wouldn't have to row reduce decimals:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

[PROF: ✓]

(c)

$$R_1 \rightarrow R_1 - 2R_3: \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_2 \rightarrow R_2 - 3R_1: \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 560]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_3 \rightarrow R_3 - 2R_1: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 560]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_2 \rightarrow (1/7)R_2: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 10 \mid 200]$$

$$R_3 \rightarrow (1/10)R_3: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 1 \mid 20]$$

Back substituting: $x_3 = 20$, then $x_2 = 80 - 2(20) = 40$, then $x_1 = -40 + 4(20) = 40$.

Checking against the copper equation, $0.5(40) + 0.2(40) + 0.4(20) = 20 + 8 + 8 = 36$.

[ANSWER: 40 kg of Alloy 1, 40 kg of Alloy 2, and 20 kg of Alloy 3]

[PROF: ✓]

[PROF: Problem 4 8/8]

Problem 5

(a)

$$\begin{array}{ccccc} [1 \ 2 \ 1] & [1 \ 2 \ 1] & [1 \ 2 \ 1] & [1 \ 2 \ 1] & [1 \ 2 \ 0] \\ [2 \ 4 \ 3] & \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] \\ [3 \ 6 \ 4] & [3 \ 6 \ 4] & [0 \ 0 \ 1] & [0 \ 0 \ 0] & [0 \ 0 \ 0] \end{array}$$

[ANSWER: RREF(A) = [1 2 0; 0 0 1; 0 0 0]]

[PROF: -1 E2 Correct RREF, but no row operation is labeled.]

(b)

$$\text{rank}(A) = 2$$

[PROF: -0.5 E7 The rank is stated with no pivot-count justification.]

(c)

Zero solutions is possible. Running the same operations on $[A \mid b]$ turns the third row

into $[0 \ 0 \ 0 \mid b_3 - b_1 - b_2]$. Whenever $b_3 \neq b_1 + b_2$, for instance $b = [0, 0, 1]^T$, that row reads $0 = 1$, which is false, so no x can satisfy the system.

Exactly one solution is impossible. Column 2 never becomes a pivot column, so x_2 is always a free variable once the system is consistent, and a free variable means more than one solution. In rank terms, $\text{rank}(A) = 2 < 3$ unknowns, so uniqueness can't occur.

Infinitely many solutions is possible. If $b_3 = b_1 + b_2$, say $b = [1, 3, 4]^T$ or $b = 0$,

the last row reads $0 = 0$ and the system is consistent, so the free variable x_2 gives

an entire line of solutions.

[PROF: ✓]

[PROF: Problem 5 5.5/7]

Problem 6

(a)

These are dependent because there are three vectors and v_3 has a zero entry, so they can't fill $\mathbb{R}^3$.

[ANSWER: dependent; span is a plane]

[PROF: -2 E7 A zero entry and a count of vectors justify nothing; the dependence needs row reduction or the relation $v_3 = v_2 - 2v_1$.]

(b)

Three vectors in $\mathbb{R}^3$ make a square matrix, so I can test independence by determinant:

$$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

Expanding along the first row with cofactors,

$$\det = 1(1 \cdot 1 - 0 \cdot 1) - 0(1 \cdot 1 - 0 \cdot 0) + 1(1 \cdot 1 - 1 \cdot 0) = 1 - 0 + 1 = 2.$$

The determinant is $2 \neq 0$, so $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ forces $c_1 = c_2 = c_3 = 0$ and the

vectors are independent. Three independent vectors in $\mathbb{R}^3$ have to span all of $\mathbb{R}^3$.

[ANSWER: independent; span is all of $\mathbb{R}^3$]

[PROF: ✓]

Grader summary

Problem 1: 5/5

Problem 2: 6/6

Problem 3: 2.5/6

Problem 4: 8/8

Problem 5: 5.5/7

Problem 6: 6/8

Total: 33/40

Problem 3 is where this paper fell apart: 3(a) gives the answer with no elimination at all, and in 3(b) you drop the zero row, set $z = 0$, and call one particular solution the solution. A free variable means infinitely many solutions, and the general solution must be written in parametric form with the parameter and its domain. Label the row operations in 5(a), and justify your claims: 'v3 has a zero entry' is not an argument for dependence, and a bare rank value earns half credit.